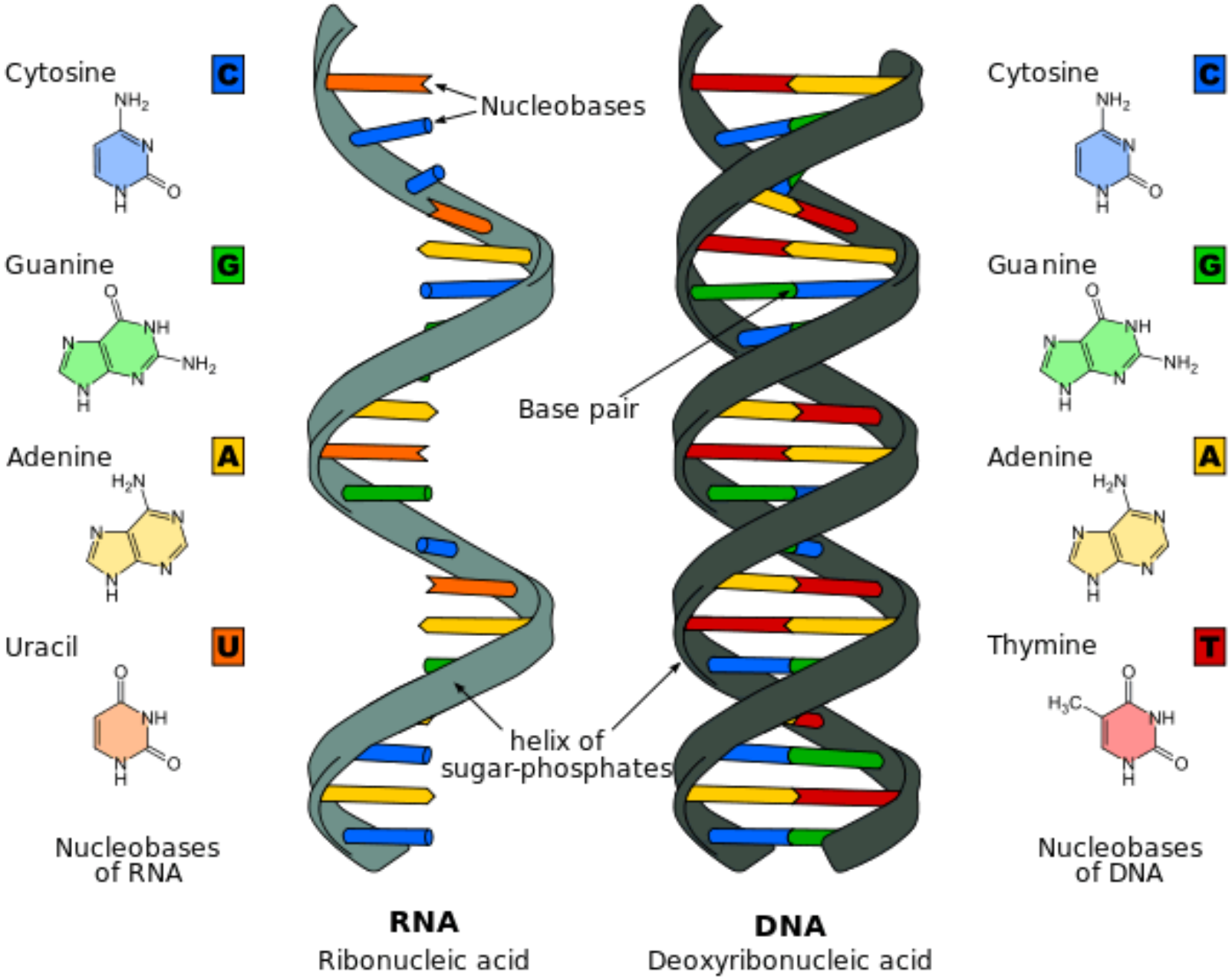
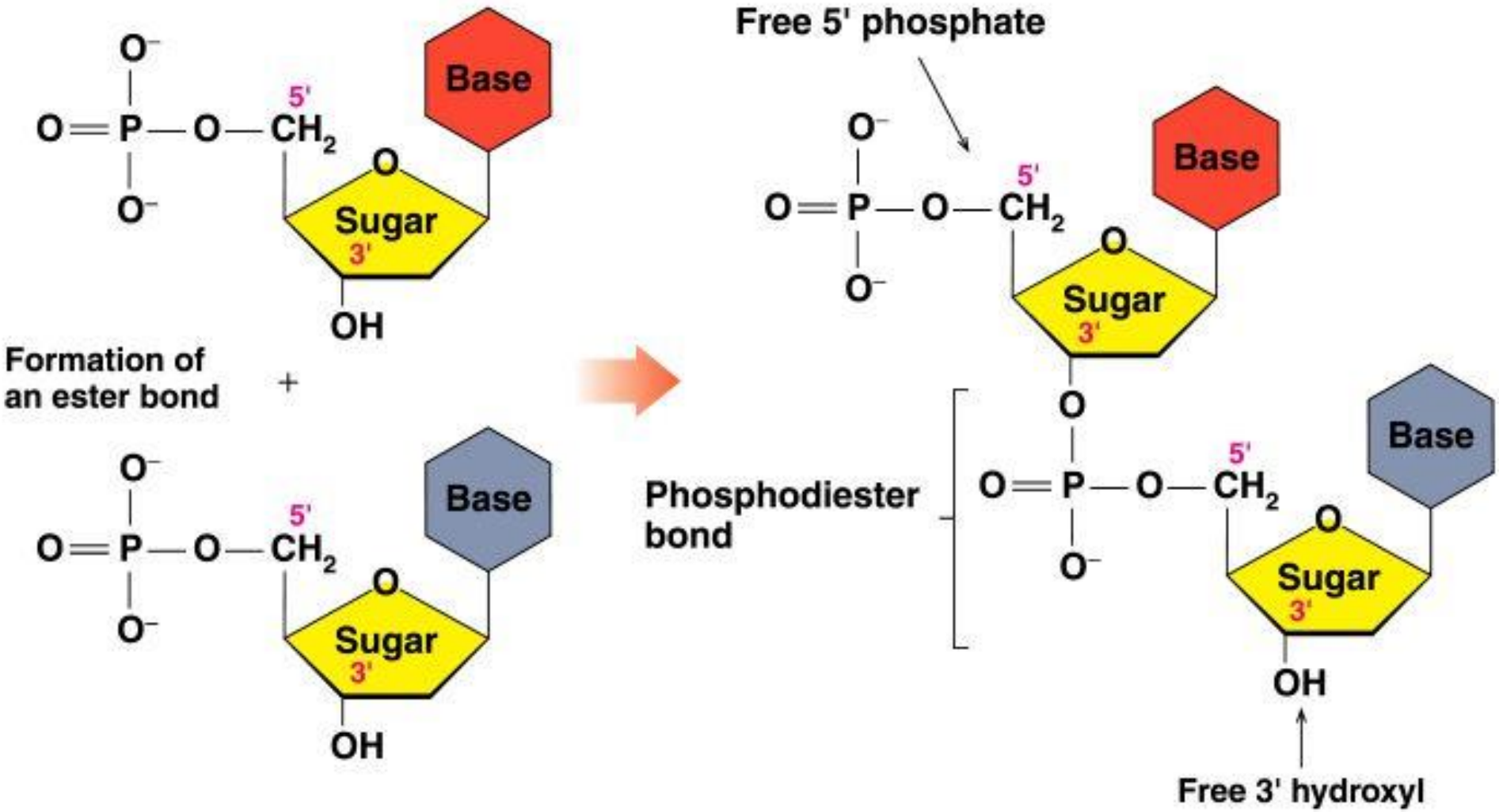
Bio-macromolecules: Molecules of Life

	Protein	Nucleic Acid	Lipid	Carbohydrates	Small molecules
Made of	Long chains of 20 kinds of amino acids	Long & short chains of nucleic acid bases form DNA, RNA, also ATP, GTP, ...	Phosphate or or other charged "head" with long hydro-carbon tail	Long & short chains of sugar molecules like glucose, fructose.	Various molecules such as hormones, vitamins, neurotransmitters, porphyrins
Functions	Most of the structure and function of living things; enzymes	Information storage (RNA, DNA), structure, enzymes, energy transfer	Energy storage Insulation Cushioning Membranes	Energy source Energy storage Structure	Mostly sending signals

Nucleic Acids

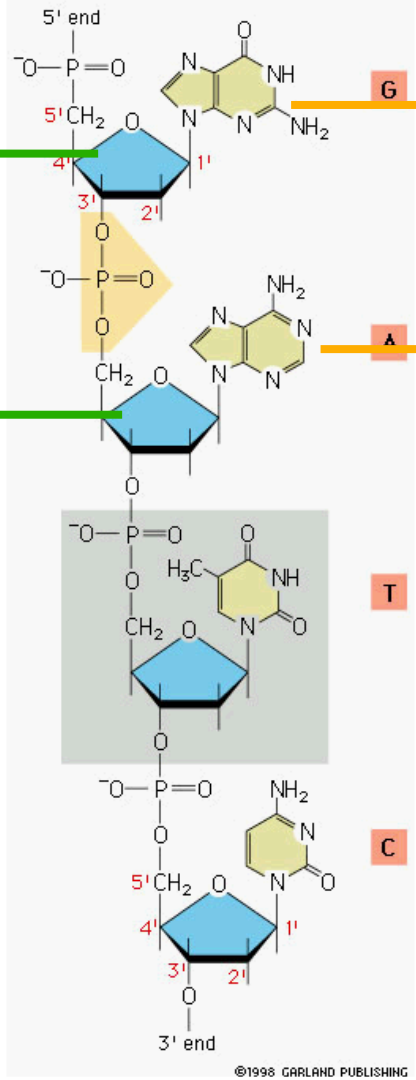


Nucleic Acid Structure Polymerization



Nucleic Acid Structure Polymerization

Sugar Phosphate "backbone"



Nucleotide

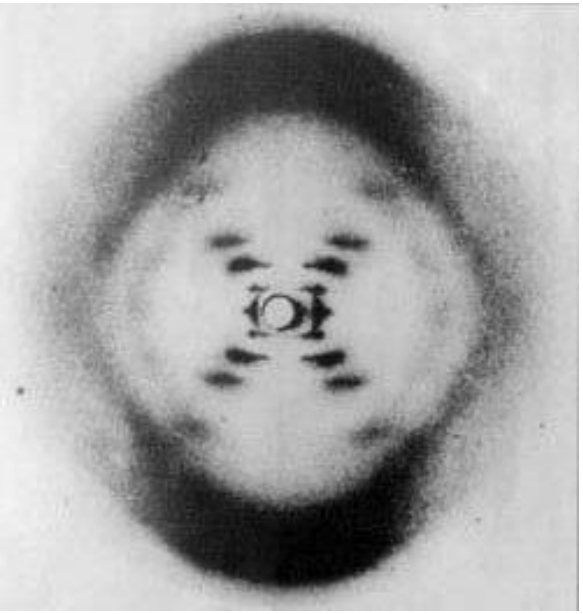
PERIPHERY OF DNA

- SUGAR-PHOSPHATE CHAINS

CORE OF DNA

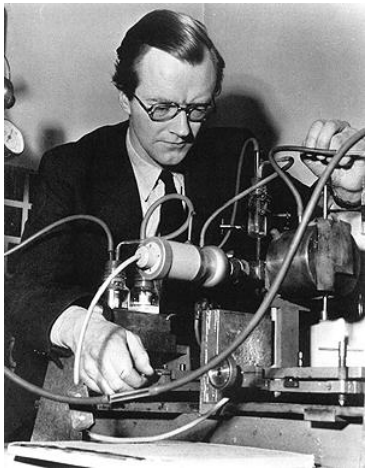
- BASES ARE STACKED IN PARALLEL FASHION
- CHARGAFF'S RULES
  - A = T
  - G = C
- "COMPLEMENTARY" BASE-PAIRING

X-Ray Diffraction Pattern of DNA Fibers

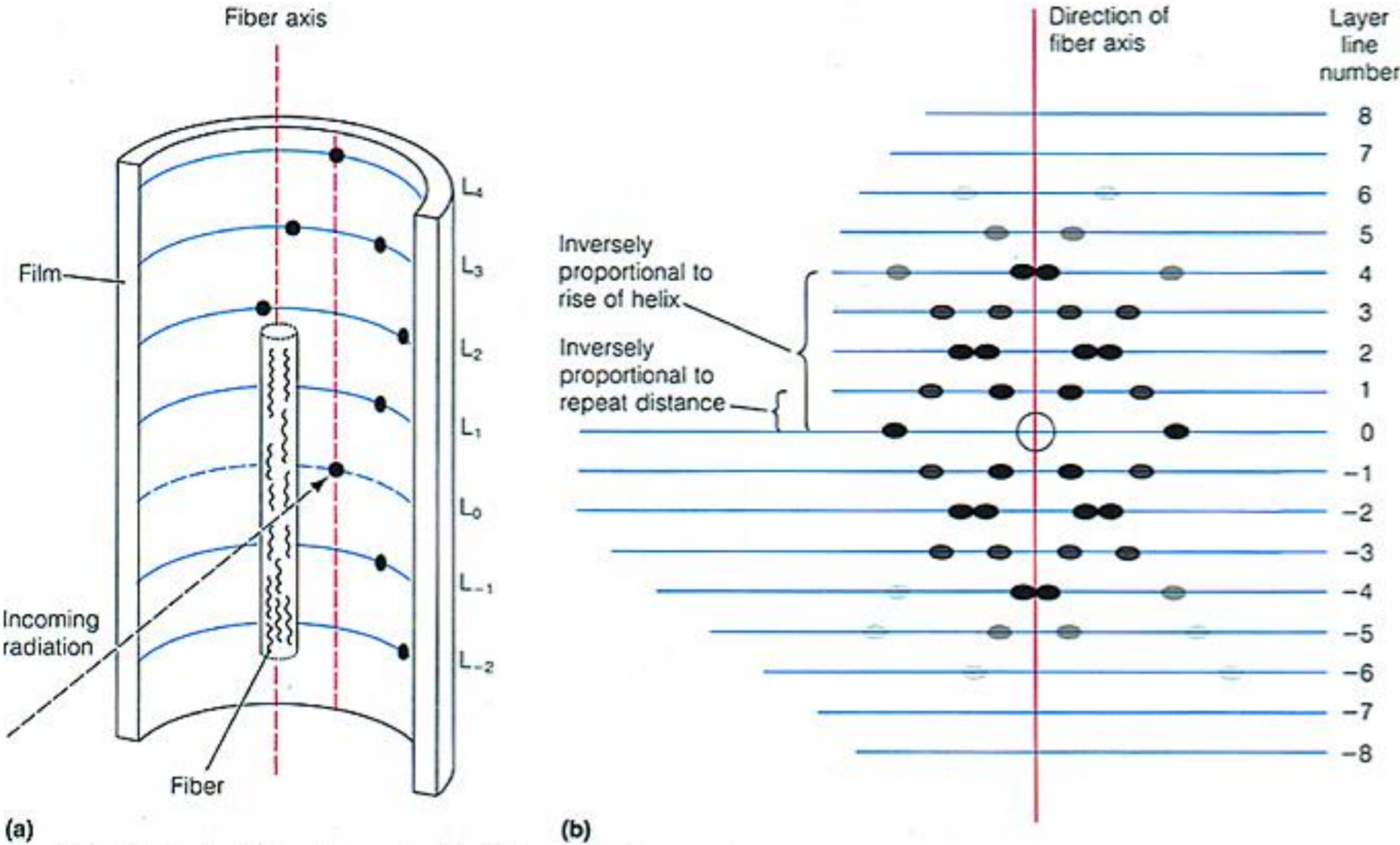


Franklin and Wilkins, 1953

Rosalind Franklin



Maurice Wilkins



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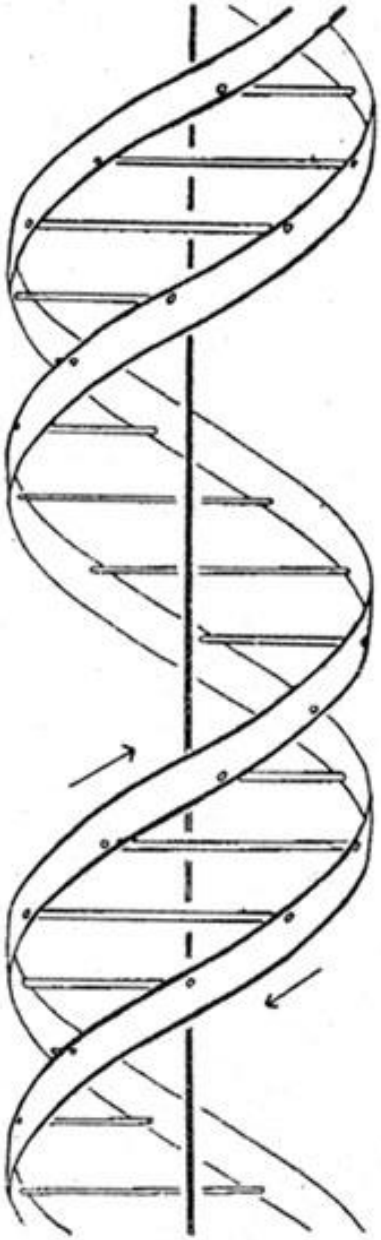

April 25, 1953

MOLECULAR STRUCTURE OF NUCLEIC ACIDS

A Structure for Deoxyribose Nucleic Acid

We wish to suggest a structure for the salt of deoxyribose nucleic acid (D.N.A.). This structure has novel features which are of considerable biological interest.

It has not escaped our notice that the specific pairing we have postulated immediately suggests a possible copying mechanism for the genetic material.

J. D. WATSON F. H. C. CRICK

Medical Research Council Unit for the Study of Molecular Structure of Biological Systems, Cavendish Laboratory, Cambridge.

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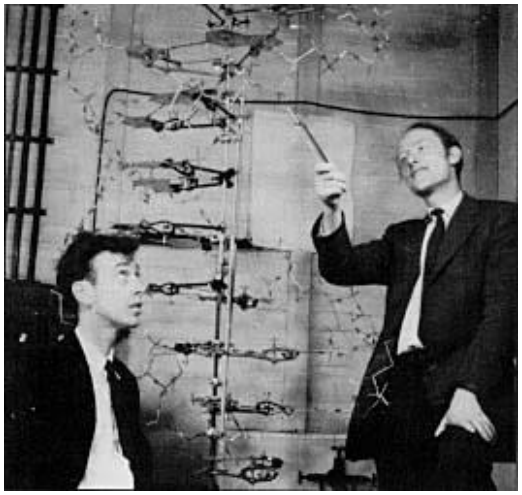
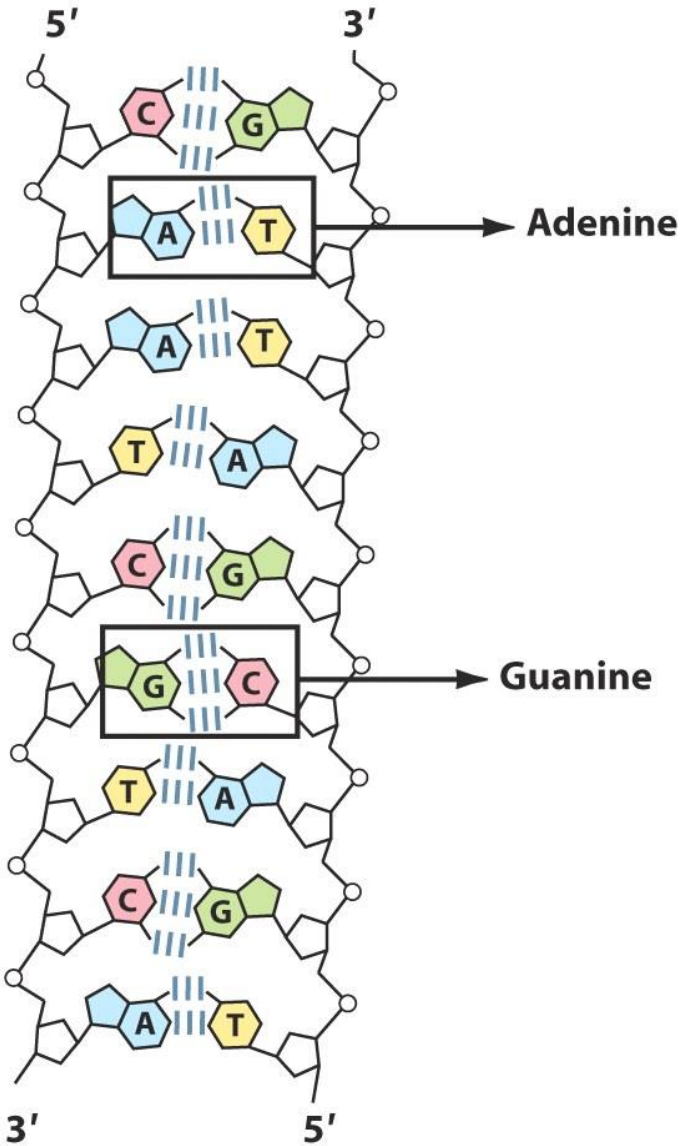
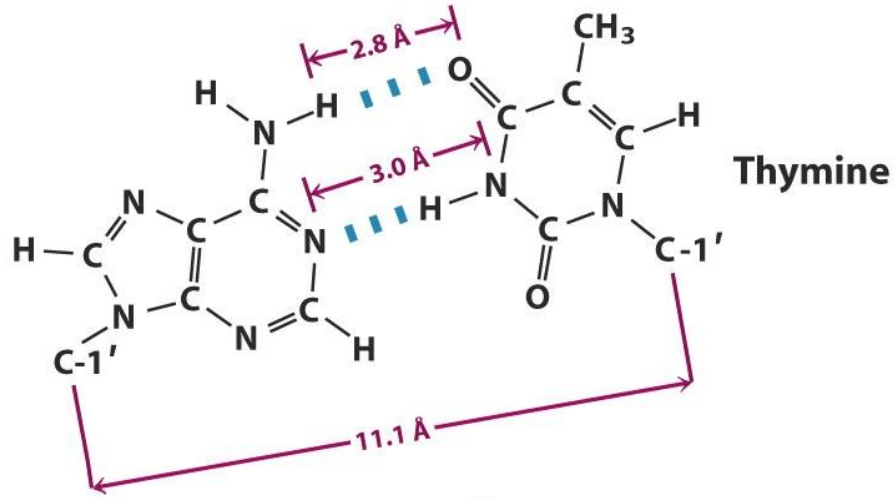
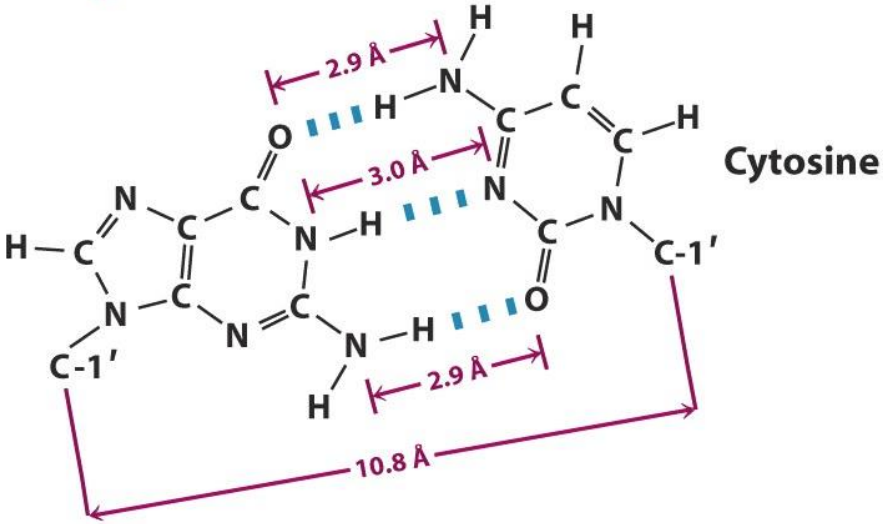


Elements of Structure in DNA Double Helix

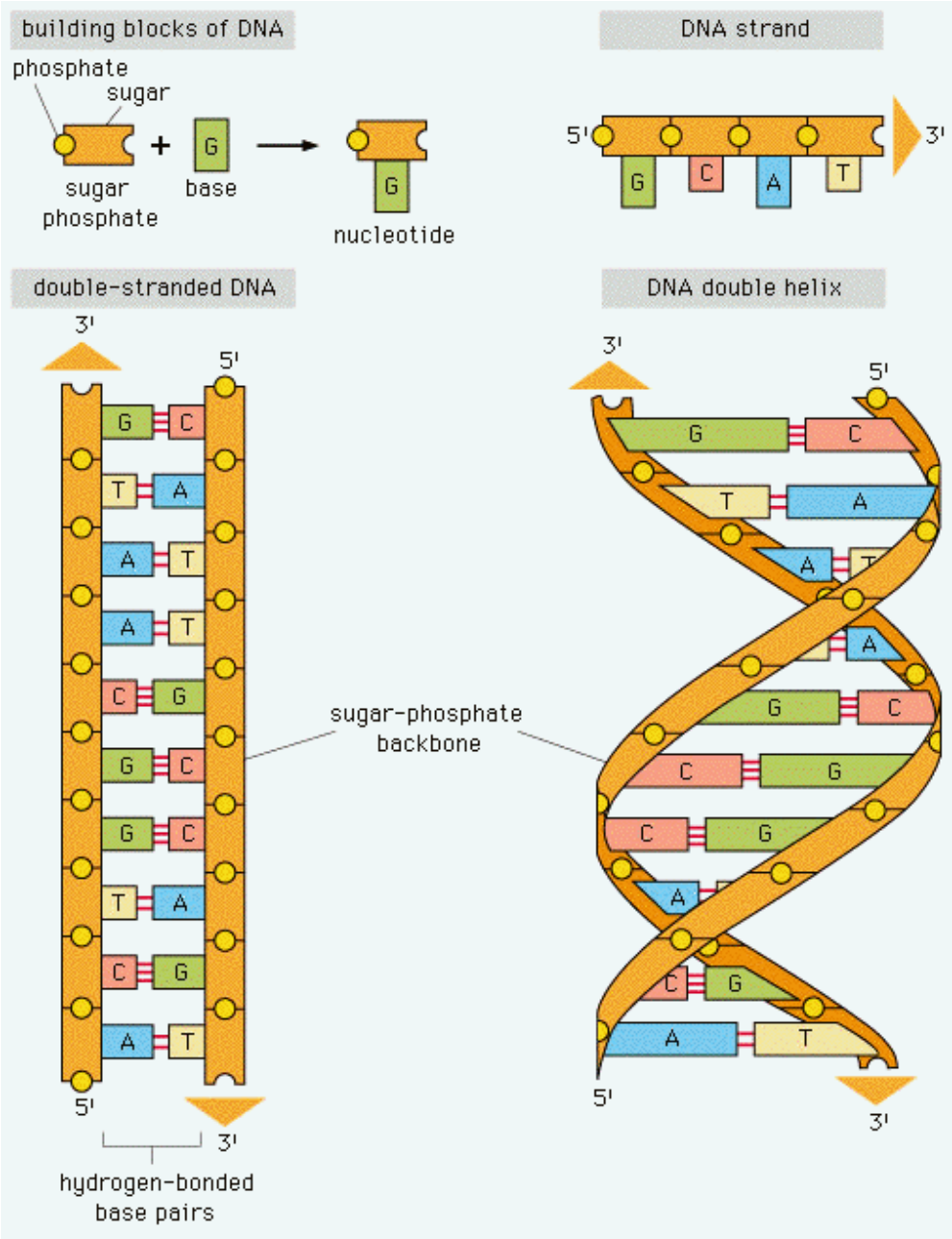


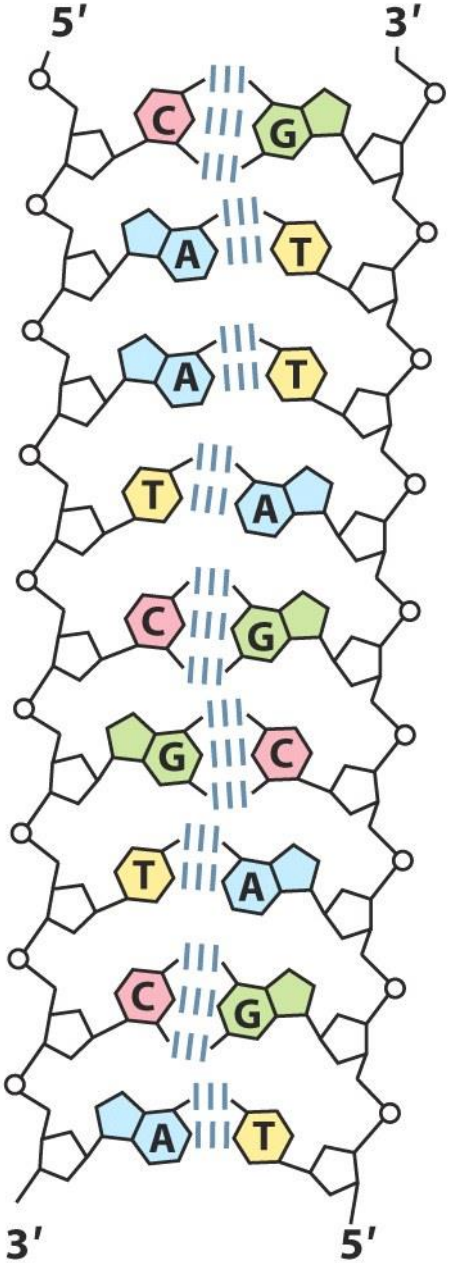




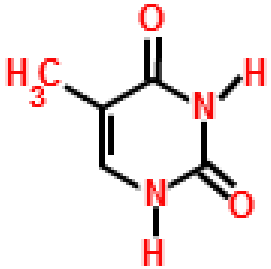
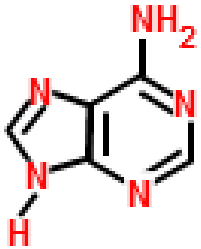
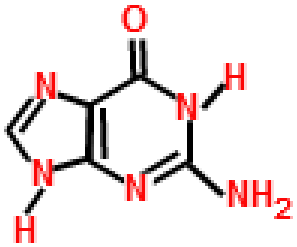
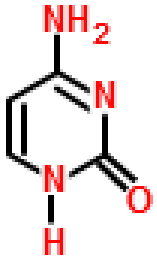
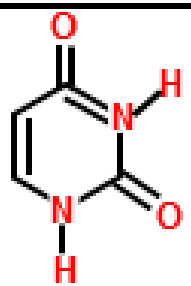
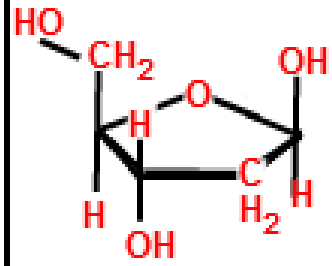
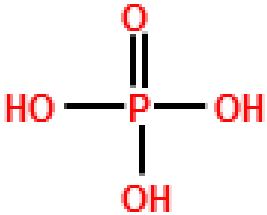
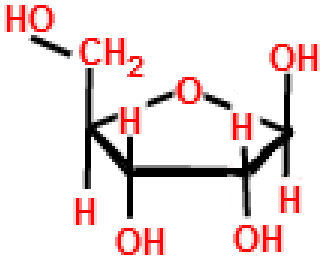
Watson-Crick Base Pairs

Elements of Structure in DNA Double Helix



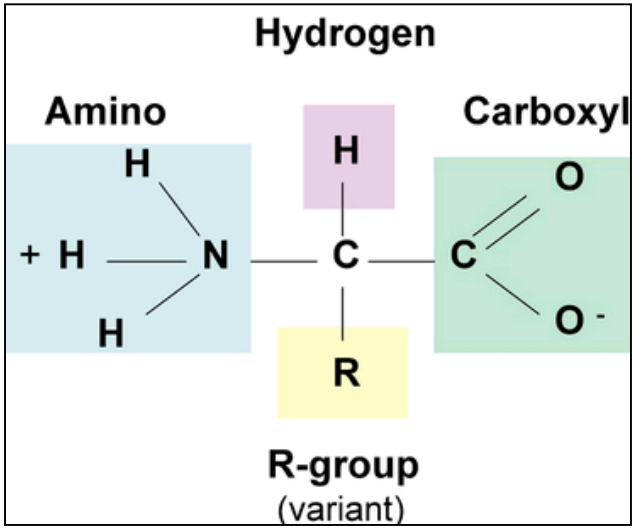


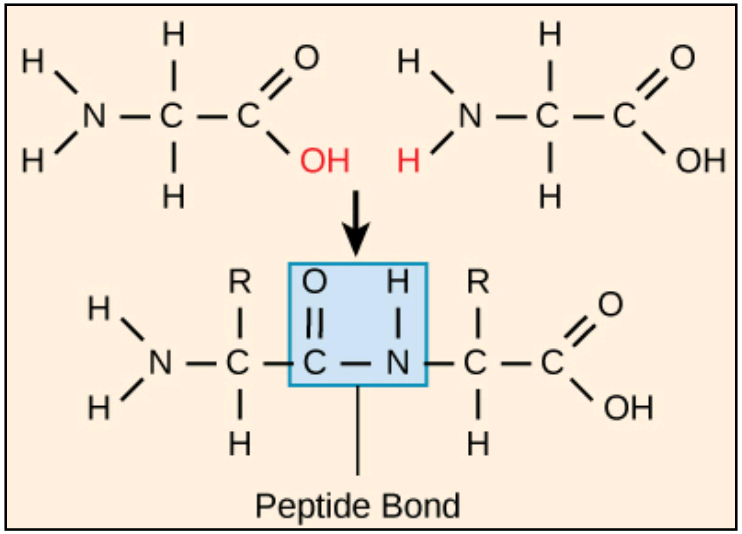
Components of Nucleic Acids

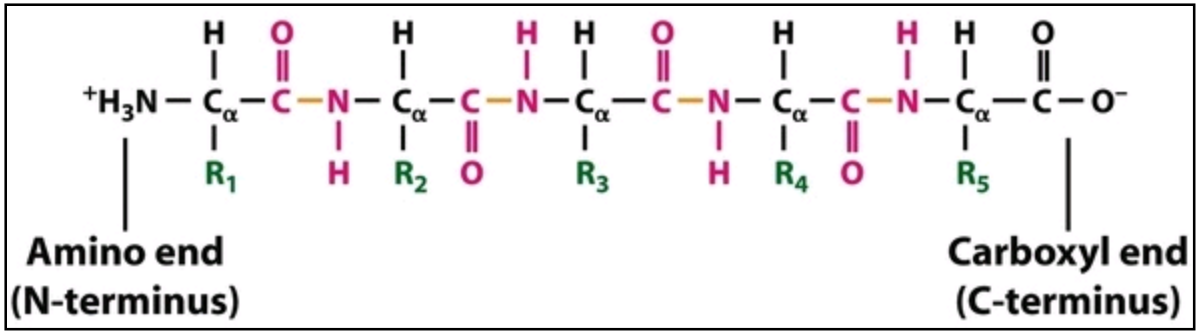
	DNA only	DNA & RNA			RNA only
Nitrogen bases	 Thymine	 Adenine	 Guanine	 Cytosine	 Uracil
sugar & phosphate	 2-Deoxyribose	 Phosphate			 Ribose

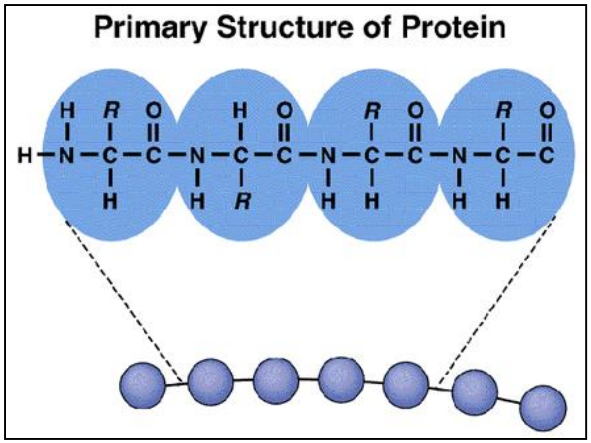
	DNA	RNA
Pentose sugar	Deoxyribose	Ribose
Base Composition	Adenine (A) Guanine (G) Cytosine (C) **Thymine (T)**	Adenine (A) Guanine (G) Cytosine (C) **Uracil (U)**
Number of strands	Double stranded (forms a double helix)	Single stranded

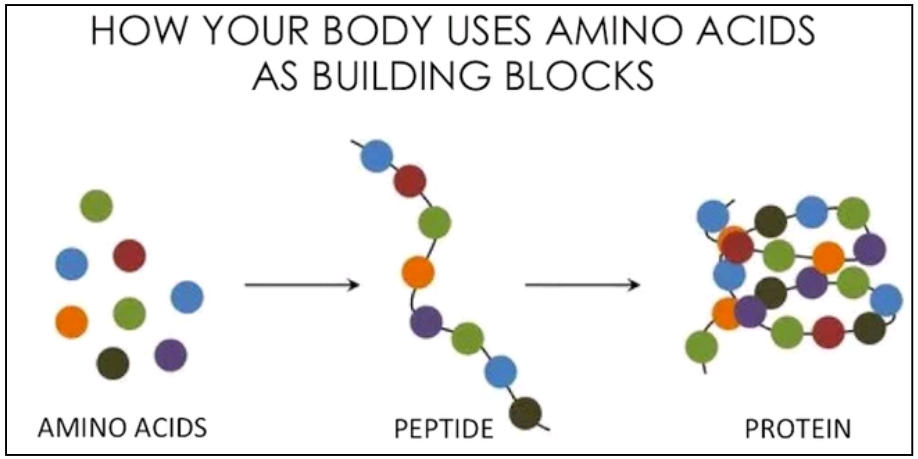
Proteins are made up of 20 amino acids linked by peptide bonds



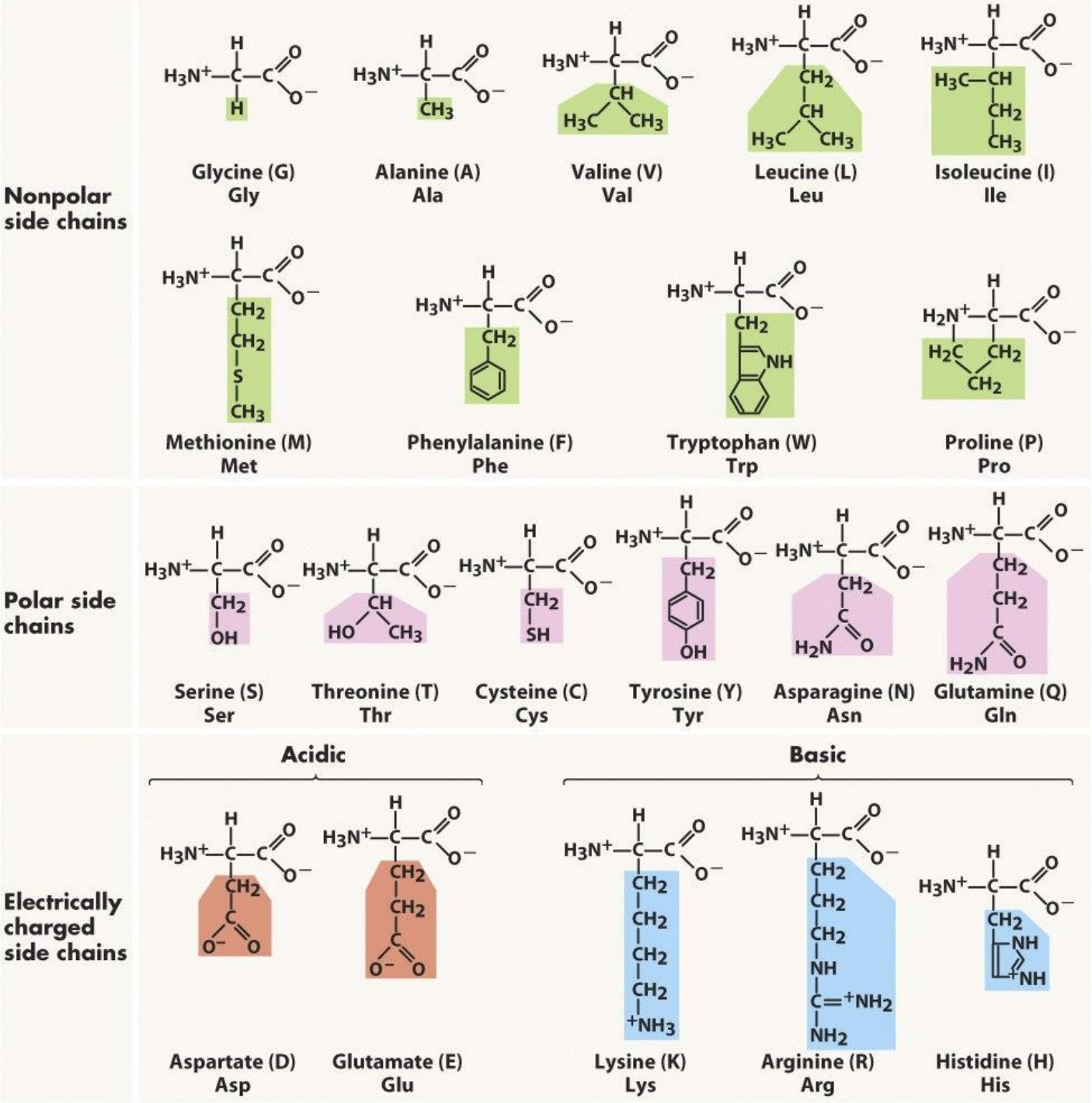




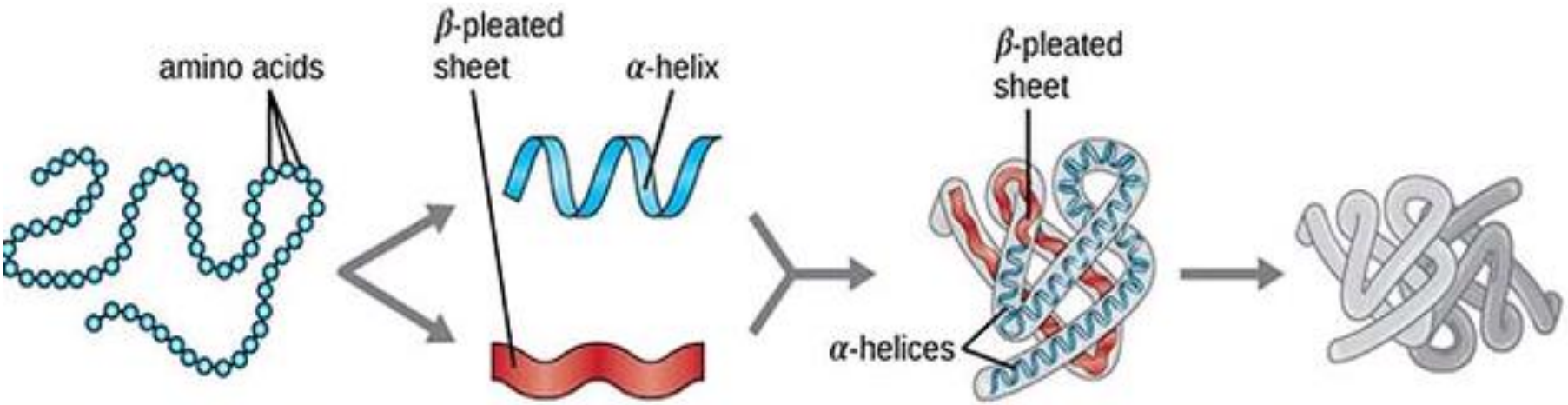




The structure of the R-group determines the chemical properties of the amino acid



Protein folding: Levels of protein structure



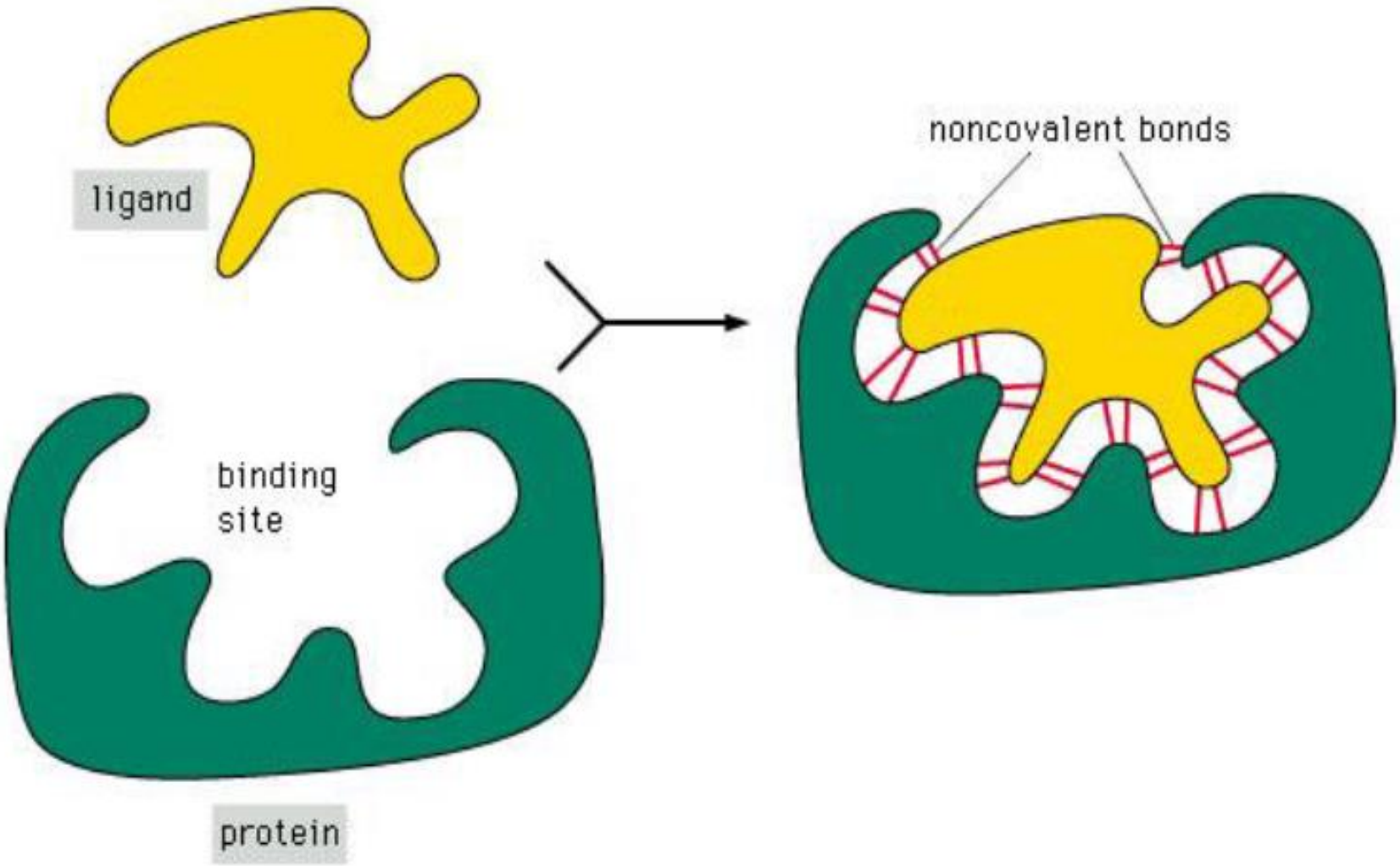
Primary Protein Structure
Sequence of a chain of amino acids

Secondary Protein Structure
Local folding of the polypeptide chain into helices or sheets

Tertiary Protein Structure
three-dimensional folding pattern of a protein due to side chain interactions

Quaternary Protein Structure
protein consisting of more than one amino acid chain

Ligand Binding



Proteins classified by Function

- **CATALYTIC:** enzymes
- **STORAGE:** ovalbumen (in eggs), casein (in milk), zein (in maize)
- **TRANSPORT:** haemoglobin
- **COMMUNICATION:** hormones (eg insulin) and neurotransmitters
- **CONTRACTILE:** actin, myosin, dynein (in microtubules)
- **PROTECTIVE:** Immunoglobulin, fibrinogen, blood clotting factors
- **TOXINS:** snake venom
- **STRUCTURAL:** cell membrane proteins, keratin (hair), collagen

Carbohydrates

- **Cells use carbohydrates for energy and structural materials**
- **Carbohydrates**
  - **Molecules that consist primarily of carbon, hydrogen, and oxygen atoms in a 1:2:1 ratio**

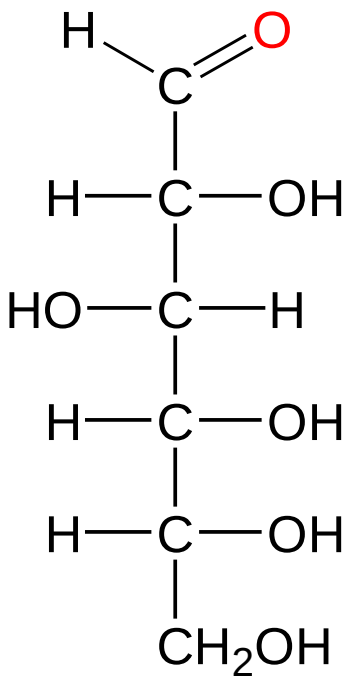
Carbohydrates (glycans) have the following basic composition:

$$(CH_2O)_n \quad \text{or} \quad \begin{array}{c} | \\ H - C - OH \\ | \end{array}$$

- **Monosaccharides - simple sugars with multiple OH groups. Based on number of carbons (3, 4, 5, 6), a monosaccharide is a triose, tetrose, pentose or hexose. 1:2:1 ratio of C:H:O- $C_6H_{12}O_6$**
- **Disaccharides - 2 monosaccharides covalently linked.**
- **Oligosaccharides - a few monosaccharides covalently linked.**
- **Polysaccharides - polymers consisting of chains of monosaccharide or disaccharide units.**

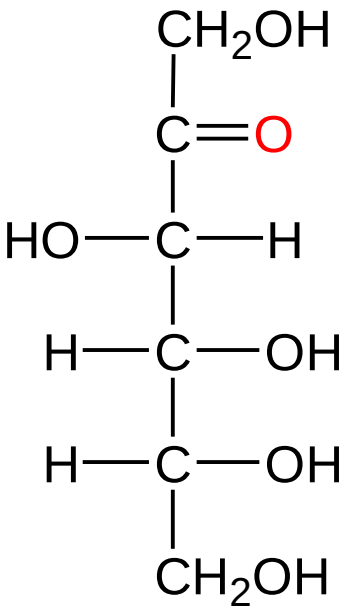
Monosaccharides

Aldoses (e.g., glucose) have an **aldehyde** group at one end.



D-glucose

Ketoses (e.g., fructose) have a **keto** group, usually at C2.

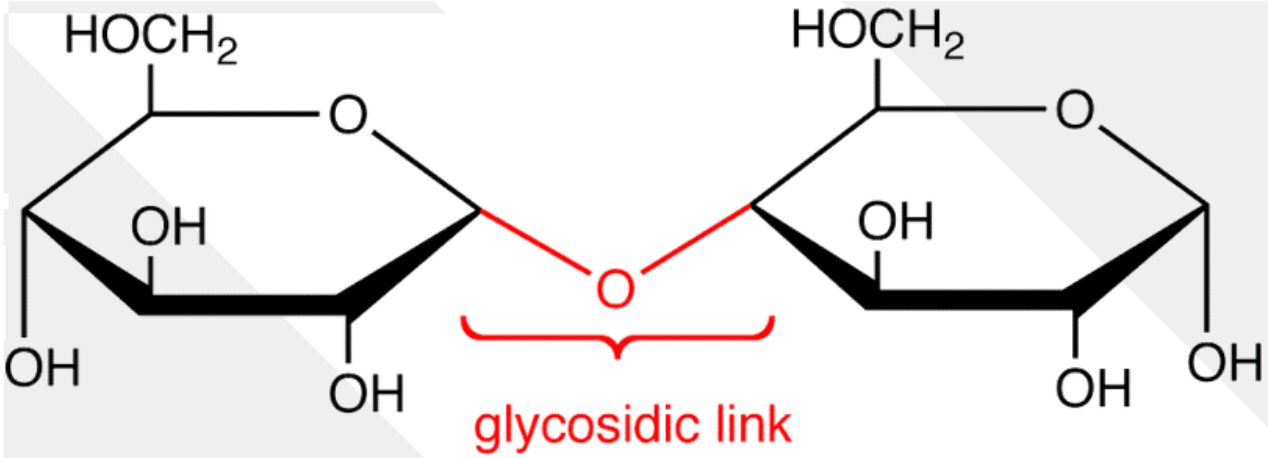


D-fructose

Most naturally occurring sugars are D isomers.

Disaccharides

- 2 monosaccharides linked together
- Key disaccharides you need to know:
  - Fructose + glucose → sucrose + H_2O
  - Glucose + galactose → lactose + H_2O
  - Glucose + glucose → maltose + H_2O



Glycosidic linkage:

The bond between monosaccharides

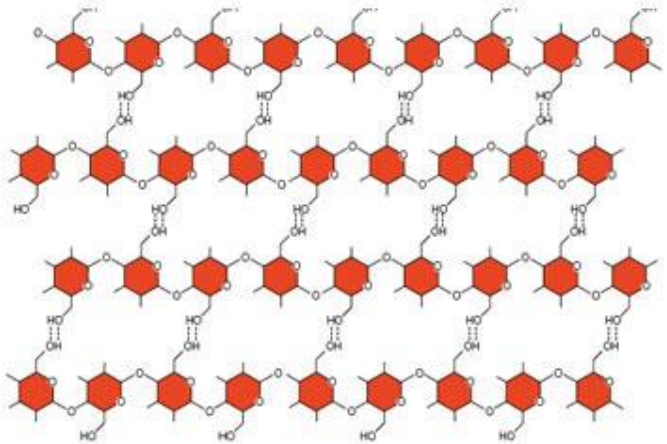
Complex Carbohydrates

Enzymes assemble complex carbohydrates (polysaccharides) from simple carbohydrate (sugar) subunits

Glucose monomers can bond in different patterns to form different complex carbohydrates

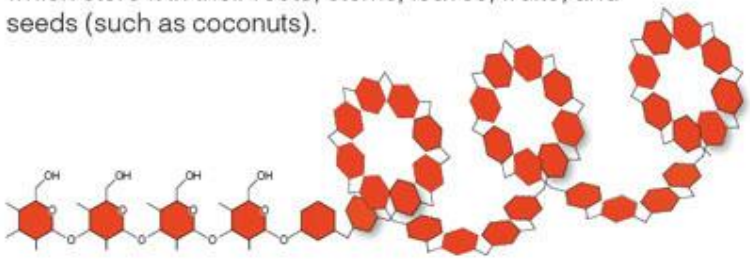
- **Cellulose (a structural component of plants)**
- **Starch (main energy reserve in plants)**
- **Glycogen (energy reserve in animals)**

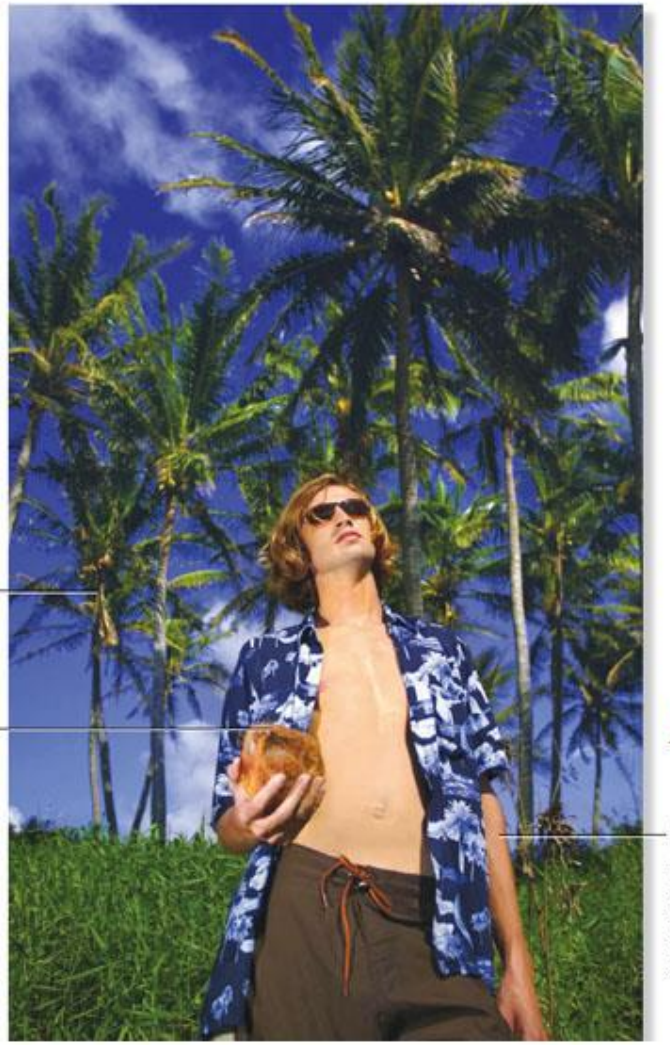
Some Complex Carbohydrates

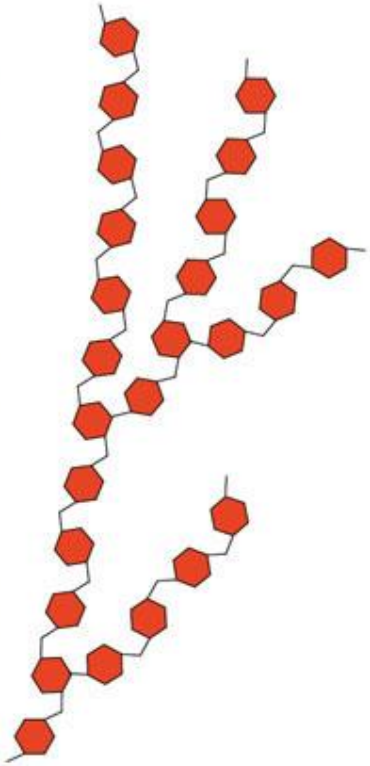


A Cellulose, a structural component of plants. Chains of glucose units stretch side by side and hydrogen-bond at many —OH groups. The hydrogen bonds stabilize the chains in tight bundles that form long fibers. Very few types of organisms can digest this tough, insoluble material.

B In starch, a series of glucose units form a chain that coils. Starch is the main energy reserve in plants, which store it in their roots, stems, leaves, fruits, and seeds (such as coconuts).







C Glycogen. In animals, this polysaccharide functions as an energy reservoir. It is especially abundant in the liver and muscles of active animals, including people.

Lipids

Lipids are a class of biological molecules defined by low solubility in water and high solubility in nonpolar solvents.

As molecules that are largely hydrocarbon in nature, lipids represent highly reduced forms of carbon and, upon oxidation in metabolism, yield large amounts of energy. Lipids are thus the molecules of choice for metabolic energy storage.

Classification

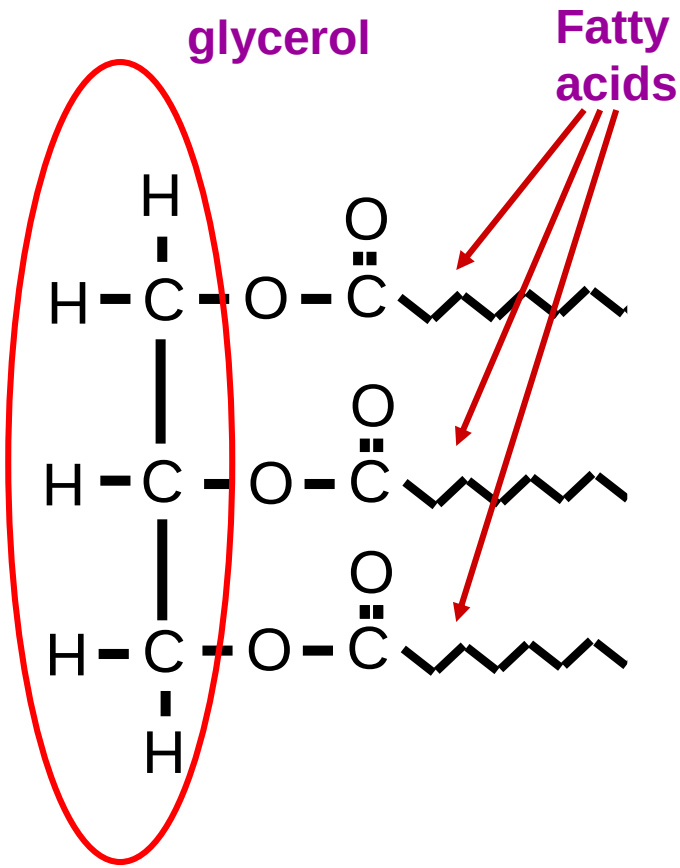
By structure:

1. **Simple: fats, oils, waxes, steroids.**
2. **Complex: phospholipids, spingolipids, glycolipids.**
3. **Derivatives: hormones, fat-solubility vitamins**

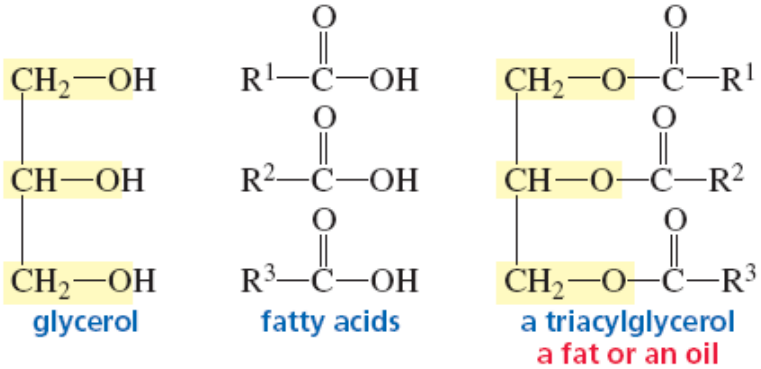
Biological functions

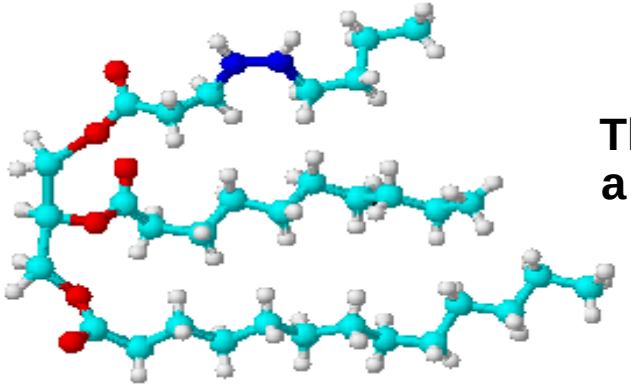
- **The most important role of lipids is as a fuel. Thus fat is the most concentrated form in which potential energy can be stored.**
- **Since fat is a bad conductor of heat, it provides excellent insulation.**
- **Fat may also provide padding to protect the internal organs.**
- **Some compounds derived from lipids are important building blocks of biologically active materials.**
- **Lipoproteins are constituents of cell walls.**
- **One more important function of dietary lipids is that of supplying the so-called essential fatty acids**

Oils and Fats



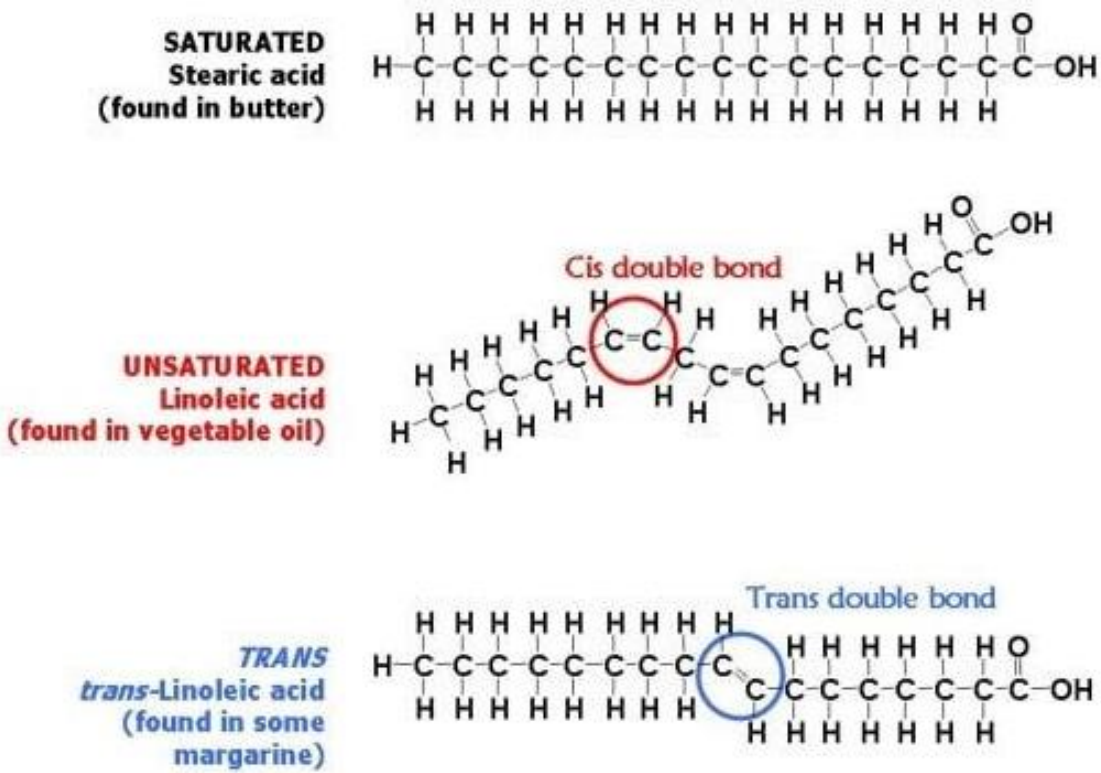
- Fats, such as triglycerides, are the most abundant source of energy in vertebrates – stored in adipose tissue that insulates the body
- Fat
  - Lipid with one, two, or three fatty acid tails
- Triglyceride
  - Made up of 1 glycerol and 3 fatty acids





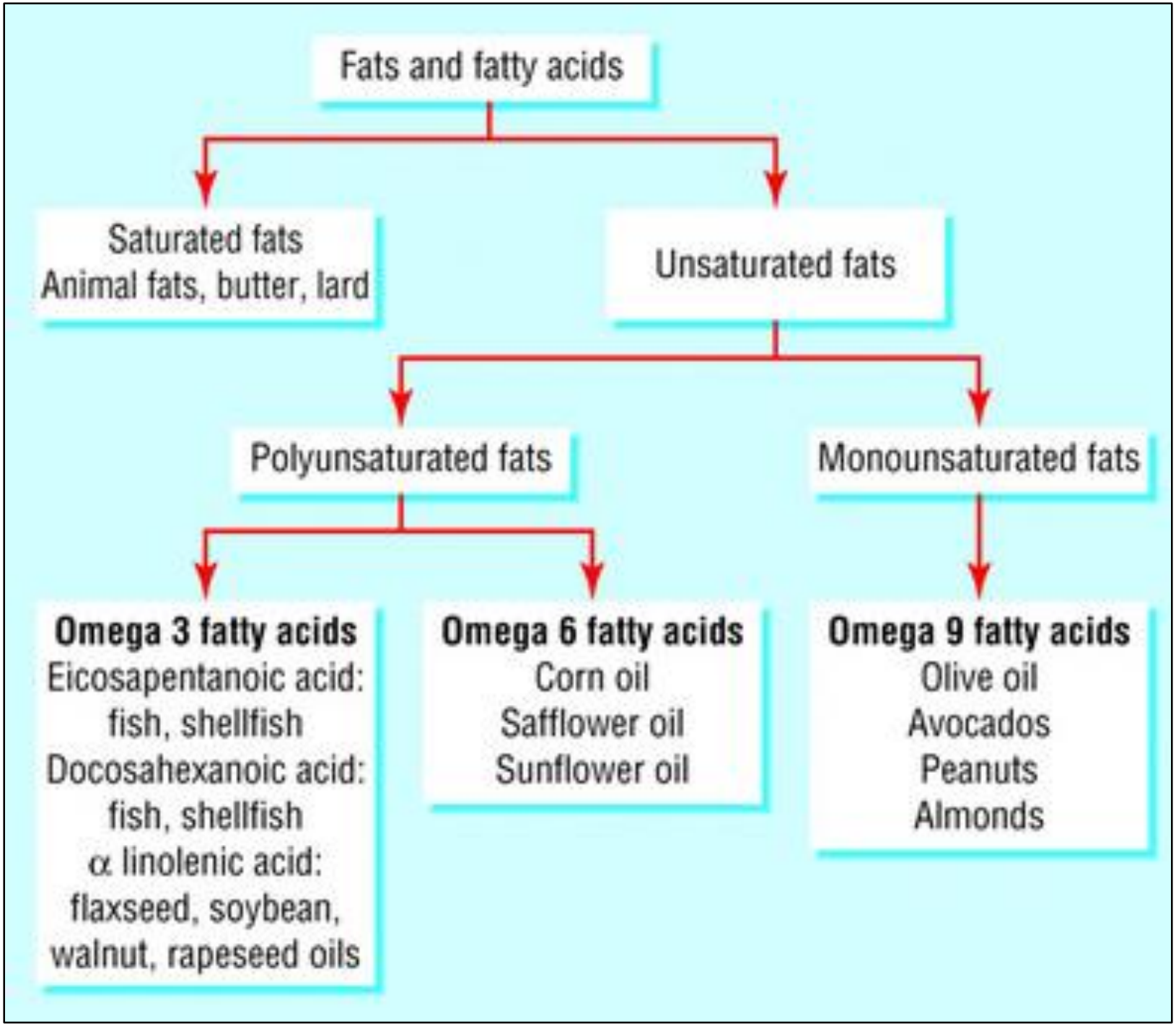
The Shape of a triglyceride is like the letter E

The Fats

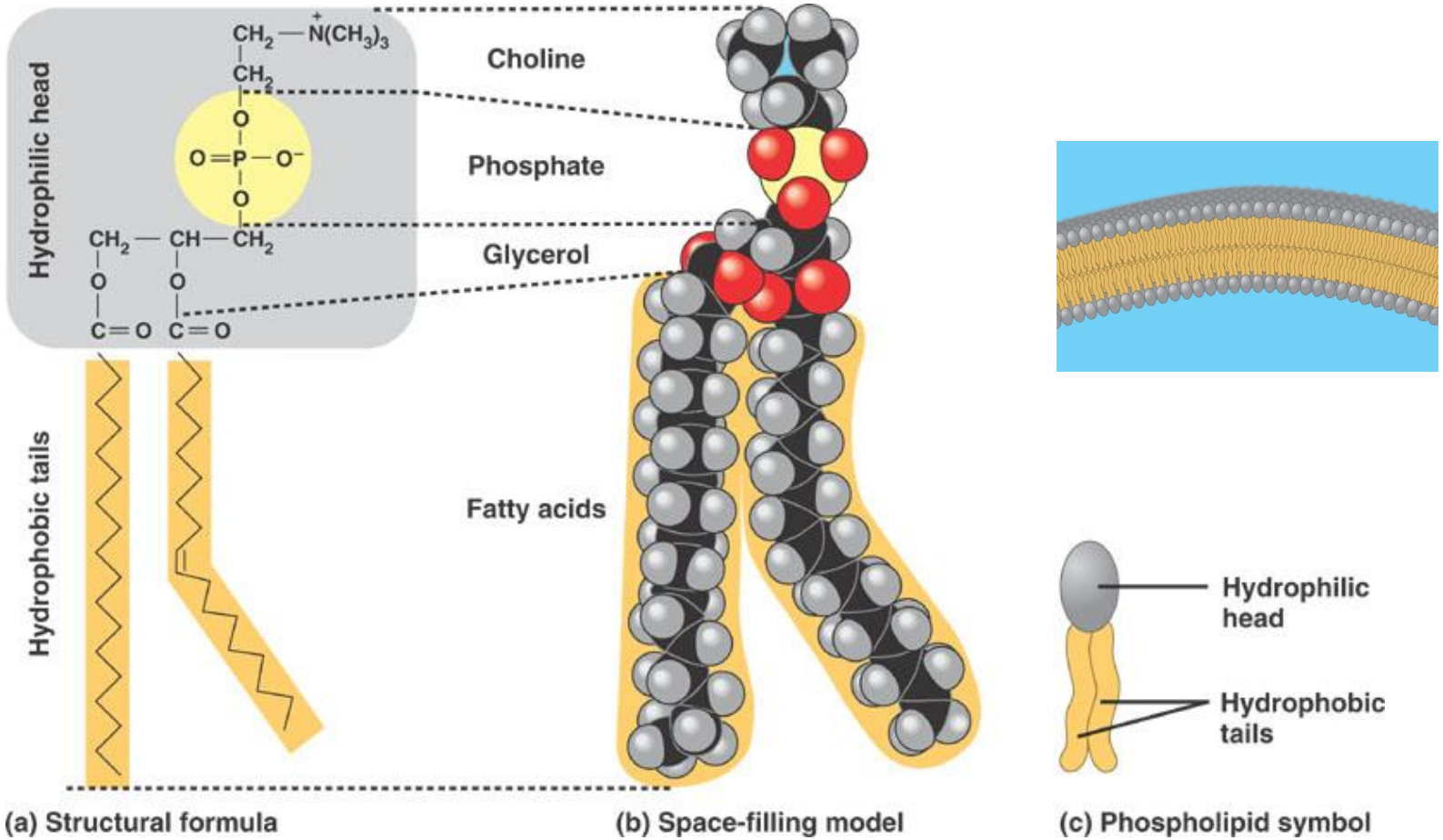


Type of Fatty Acid	Double Bonds	Diagram
Saturated	None	
Monounsaturated	One	
Polyunsaturated	Multiple (>1)	

- **Saturated fats pack more tightly than unsaturated fats, and tend to be more solid**
- **Saturated fat**
  - **Fatty acid with no double bonds in its carbon tail**
  - **Molecules are more straight**
  - **Fats: solid at room temperature**
- **Unsaturated fat**
  - **Lipid with one or more double bonds in a fatty acid tail**
  - **Causes "kinks" in molecule**
  - **Liquid at room temperature**

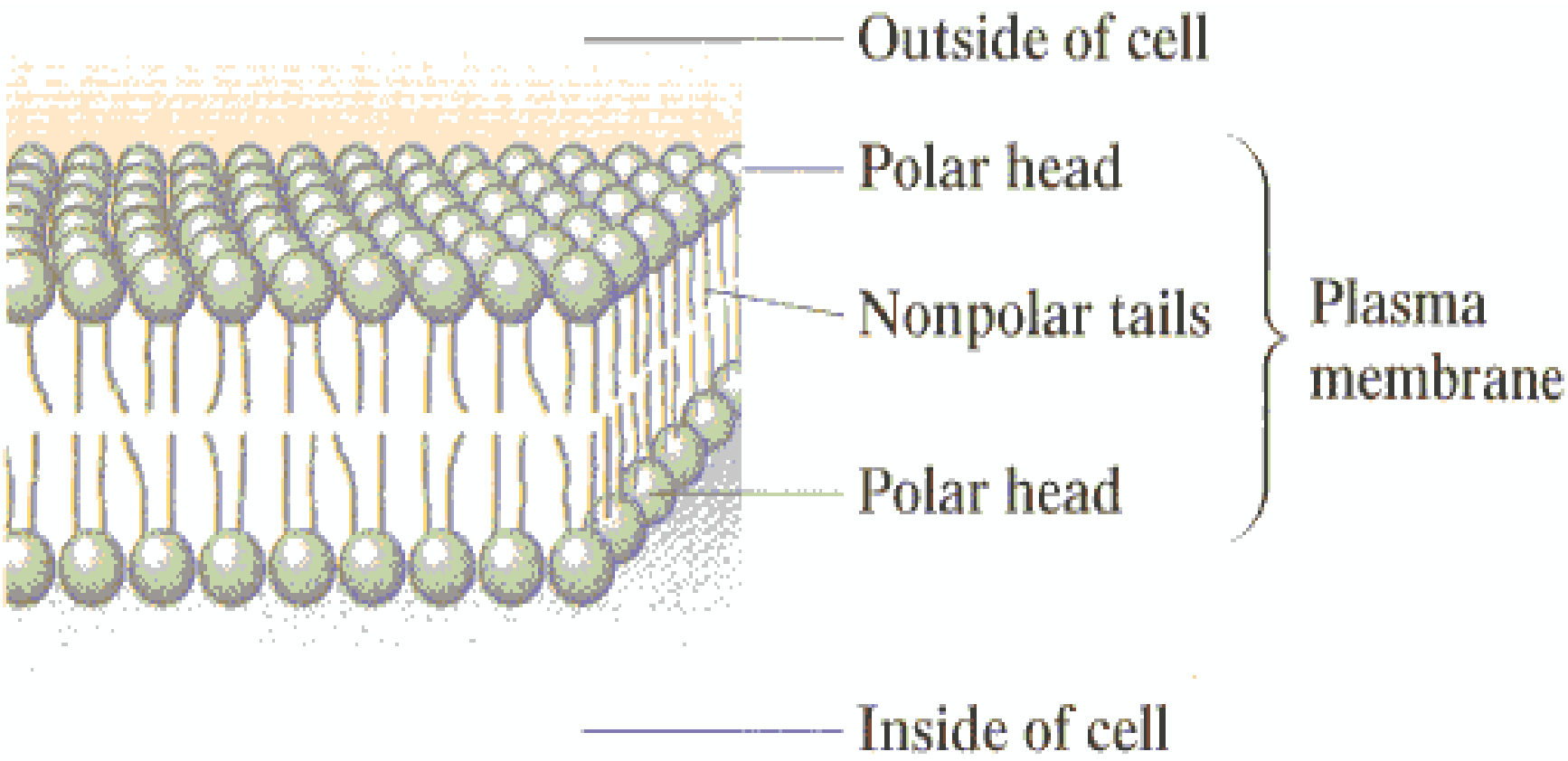

Fats and fatty acids
Saturated fats
Animal fats, butter, lard
Unsaturated fats
Polyunsaturated fats
Monounsaturated fats
Omega 3 fatty acids
Eicosapentanoic acid:
fish, shellfish
Docosahexanoic acid:
fish, shellfish
α linolenic acid:
flaxseed, soybean,
walnut, rapeseed oils
Omega 6 fatty acids
Corn oil
Safflower oil
Sunflower oil
Omega 9 fatty acids
Olive oil
Avocados
Peanuts
Almonds

Phospholipids: main structural component of cell membranes

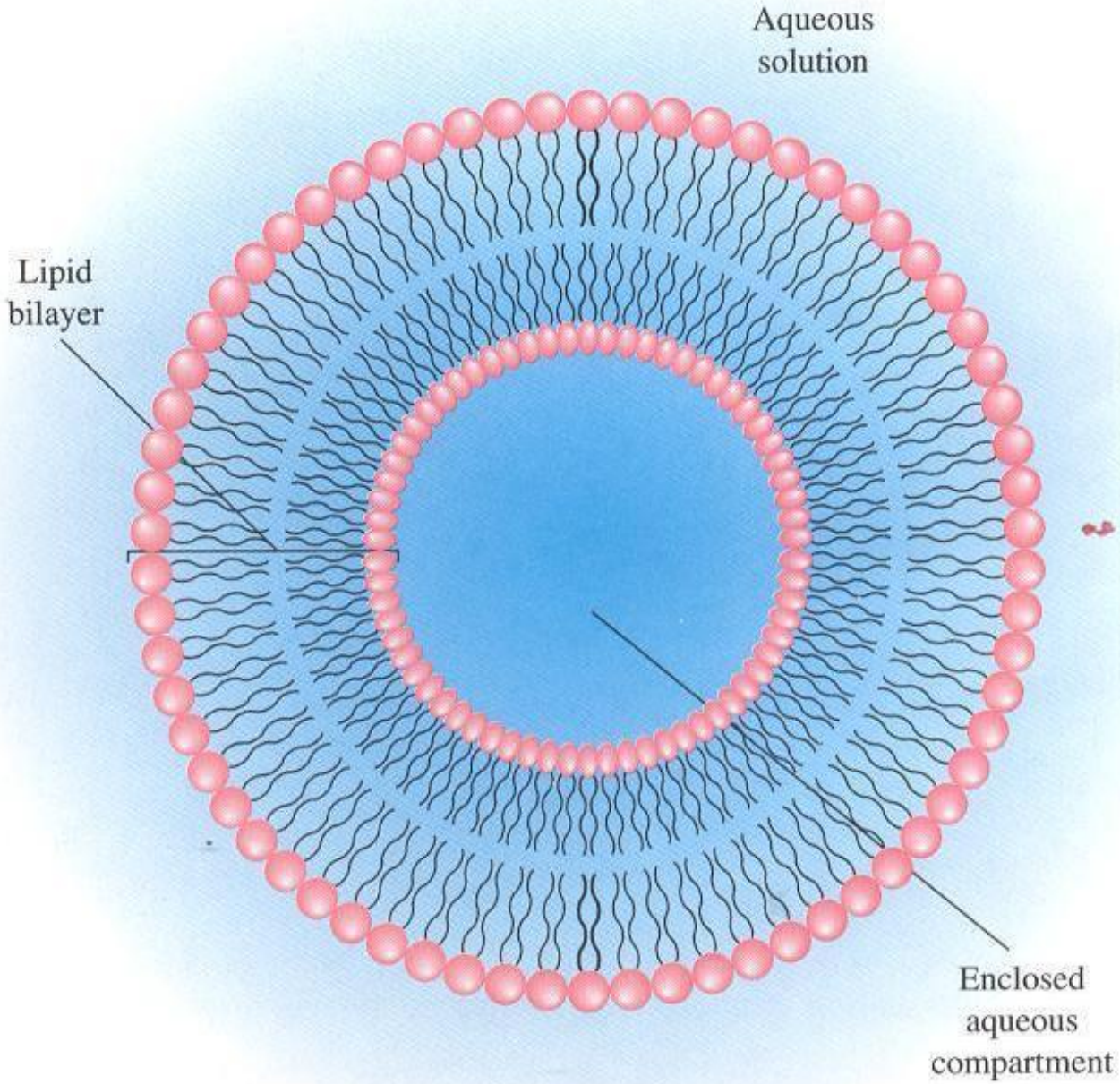


- **Structurally and chemically similar to triglycerides**
- **One fatty acid is replaced with a phosphate group**
- **Phosphate group is hydrophilic**
- **Fatty acids are hydrophobic**

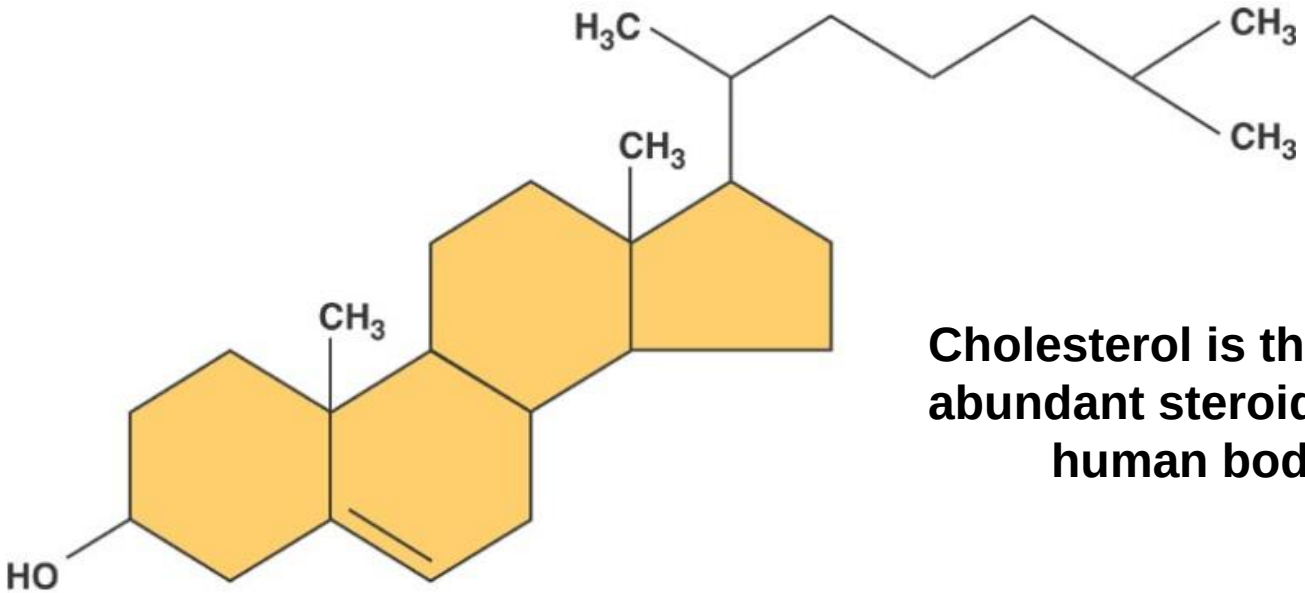
Lipid bilayer of plasma membrane



Liposome


Aqueous solution
Lipid bilayer
Enclosed aqueous compartment

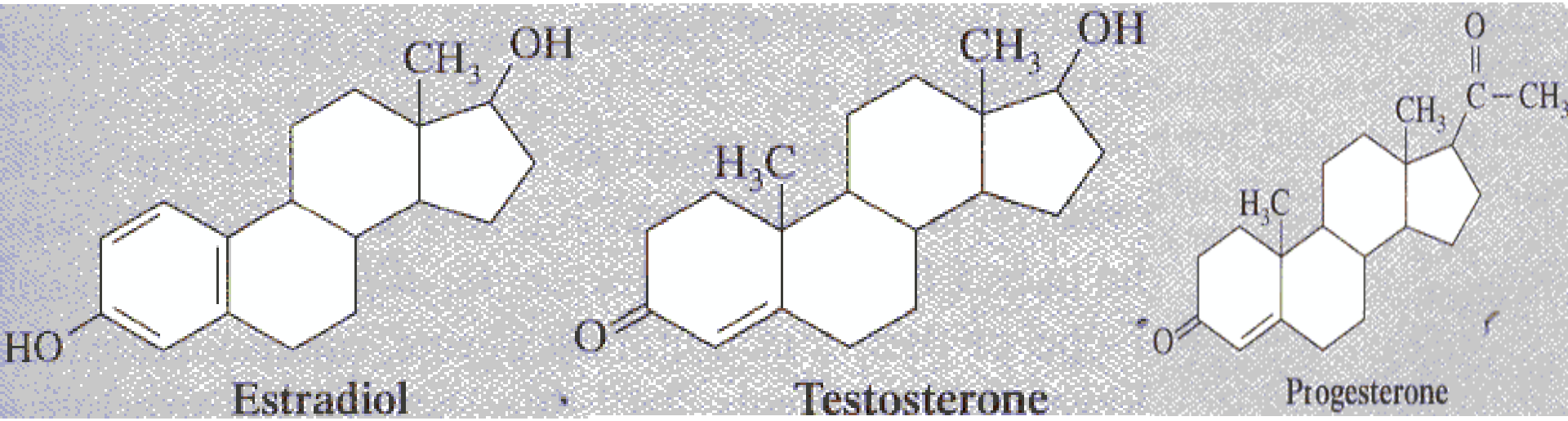
Steroids



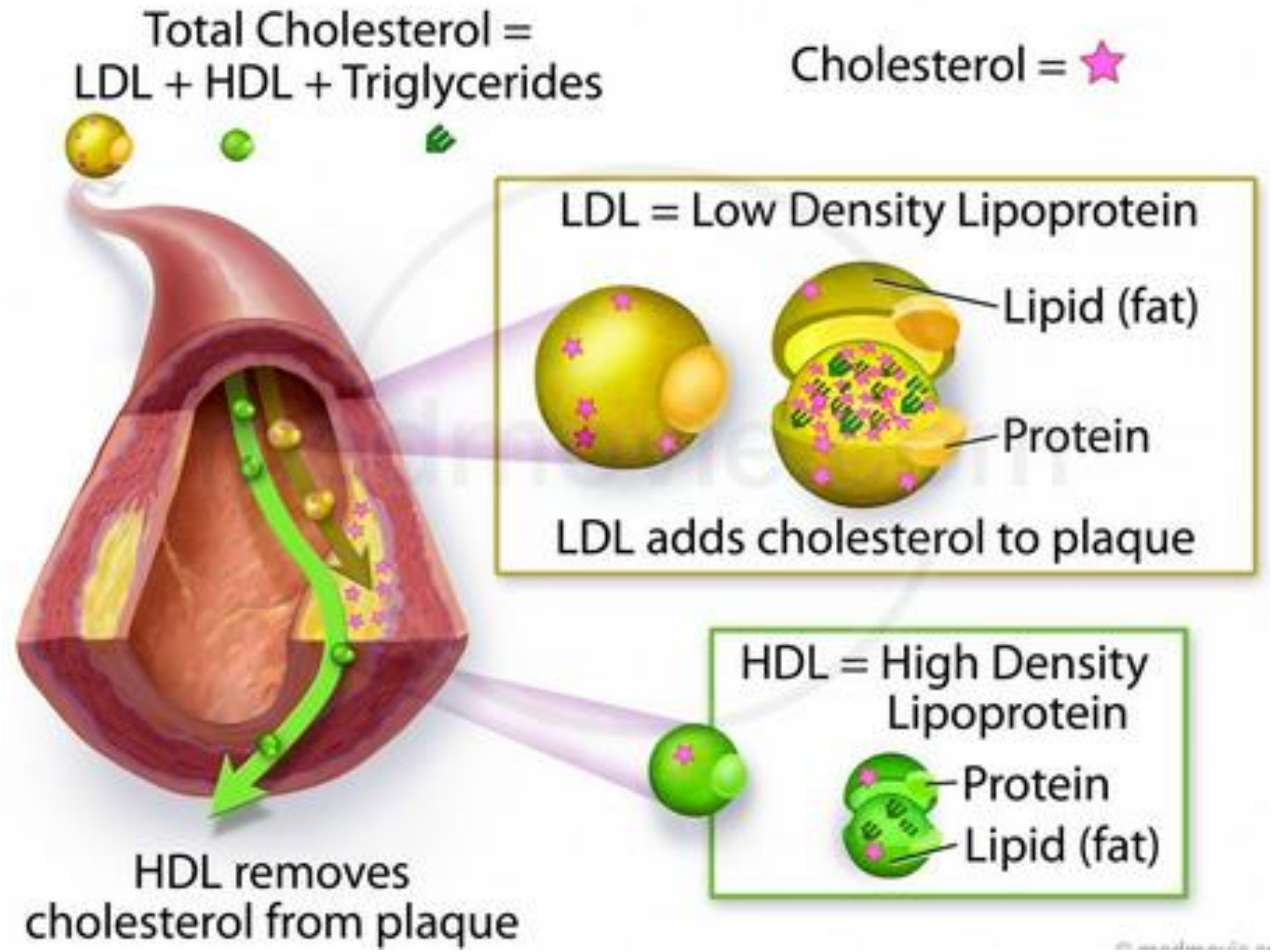
Cholesterol is the most abundant steroid in the human body

- **Chemically and structurally different from other lipid categories (no fatty acid tails)**
- **Classified as a lipid because of their nonpolar/hydrophobic nature**
- **Three of the rings are six-membered, while the fourth is five-membered.**
- **Chemical messengers (hormones)**
- **Structural components in cell membranes**

Steroids Hormones



Hormones are chemical messengers produced by ductless glands.


Total Cholesterol =
LDL + HDL + Triglycerides
Cholesterol =
LDL = Low Density Lipoprotein
Lipid (fat)
Protein
LDL adds cholesterol to plaque
HDL = High Density Lipoprotein
Protein
Lipid (fat)
HDL removes cholesterol from plaque

Normal Diet

Normal Cell

"Bad" LDL → HDL "Good"

LDL carries dietary fats into your cells

HDL carries impurities out of your cells

Diet Rich in *(CIS)* UNSATURATED FATS

Healthy Cell

LDL → HDL Raises HDL

Diets rich in **unsaturated (*cis*) fats** *lower* cholesterol in the blood

Diet Rich in SATURATED FATS

Unhealthy Cell

Raises LDL LDL → HDL

Diets rich in **saturated fats** *raise* cholesterol in the blood

Diet Rich in *TRANS* FATS

Dysfunctional Cell

Raises LDL LDL → HDL Lowers HDL

Diets rich in **trans fats** *significantly raise* cholesterol in the blood

FATS

✗ Trans Fats

- Hydrogenated vegetable oils
- Fast foods
- Cakes/pastries
- Chocolate
- Deep Fried Food

✗ Saturated Fats

Vegetable Fats

- Coconut
- Palm oil
- 3-in-1 & 2-in-1 beverages, creamer, condensed milk

Animal Fats

- Poultry skin
- Fatty meat
- Butter
- Ghee
- Tallow / lard
- Full cream dairy products

✓ Unsaturated Fats

Polyunsaturated

- Corn oil
- Soybean oil
- Sunflower oil
- Seeds
- Cold-water fish

Monounsaturated

- Olive oil
- Canola oil
- Peanut oil
- Sesame oil
- Avocado
- Most nuts